Water Treatment Plant Facility-Class Design Basis

The Governing Facility Class of the Water Sector Design Basis

DBA-W-FC1 — Facility-Class Design Basis, DBA-W Series

Issued Under: DBA-W-Sector_Framework v1.14; DBA-MA Parent Framework

Version 0.6 — DRAFT

August 2026

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Working Draft — Released for Review and Discussion
Parent Document: DBA-W-Sector_Framework v1.14

SUGGESTED CITATION

T. Roxey, "DBA-W-FC1 — Water Treatment Plant," Version 0.6, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document establishes the facility-class design basis for the water treatment plant, the first and governing facility class of the DBA-W series. At the resolution of the treatment-plant asset class, and site-neutrally, it develops the bounding events, consequence chains, design parameters, adequacy result, and dual design-basis architecture that DBA-W-Sector_Framework v1.14 establishes in general terms but does not develop for any single asset class. Its purpose is to be the class-level document from which any surface- or groundwater treatment plant, and any site-, state-, or contract-specific derivative, inherits by reference. It is written to stand on its own for an applicant and, where appropriate, to be offered for one-time regulatory review and referenced by many subsequent applications.

1.2 Scope

The facility class is the potable water treatment plant in both its surface-water and groundwater forms, together with the chemical storage and feed systems on which disinfection depends, the operational technology that governs treatment, and the finished-water interface to distribution. Distribution pump stations (FC2), storage reservoirs (FC3), seawater reverse-osmosis desalination (FC4), and wastewater treatment facilities (FC5) are separate facility classes. They are out of scope except where a treatment plant's boundary conditions depend on them. The class is defined at the level that stays true across sites: the governing function, which is the production of safe potable water through disinfection, rather than the specific unit processes of any one plant. Defining the class at the governing-function level is what lets one design basis apply across plants whose unit processes differ; the specific processes are where a site-specific derivative adds detail, not where the class is drawn. On-plant storage used to achieve disinfection contact time — clearwells and equivalent finished-water contact volume — is treated as in-plant here, part of the treatment barrier, rather than as distribution storage (FC3) or a pump-station interface (FC2).

1.3 The Reference Plants

Two reference configurations bound the class. The conventional surface-water plant — coagulation, flocculation, sedimentation, filtration, and disinfection, with bulk chemical storage — is the governing configuration, because it carries the full chemical inventory and the most complex control surface. The groundwater plant, typically disinfection-primary with limited pretreatment, is the reduced-scope variant: it inherits the class requirements that bear on disinfection and chemical handling and sheds those specific to multi-stage surface treatment. Where a requirement is stated for the class, it is stated for the governing surface-water configuration and applies to the groundwater variant to the extent the corresponding systems are present.

1.4 Relationship to Governing Documents

This instrument is issued under DBA-W-Sector_Framework v1.14 and inherits the DBA-MA parent framework. It does not repeat the sector framework's threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, or command-broker and Bright Line architecture; it inherits them, and Section 8 is the required inheritance map. It adds only what is specific to the treatment-plant class. Section references of the form “framework §N” resolve against DBA-W-Sector_Framework v1.14.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application, and the obligation it creates — is defined in FD-BL (Foundational Definition — The Bright Line), which governs. The Broker role that discharges that obligation, the Line-to-Broker pairing invariant, and the competence rule are defined in FD-BR. Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD. Two determinations bind under FD-BL: FD-BL-D1 governs the mode by which the authorization obligation may be discharged, and FD-BL-D2 governs the integrity of the basis on which an authorization is granted. FD-EV reserves the term “envelope”. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it, and adopts the water-sector conservative implementation stated in DBA-W-Sector_Framework §5.5. Instruments of the Foundational Definitions family are cited by designation and without a version token, under the citation-form determination; FD-REG, the Governing Stack Register, is authoritative for their version state and resolves each designation to the issue in force.

2. The Water Treatment Plant Facility Class

2.1 Defining Characteristics

For design-basis purposes, the water treatment plant is defined by three features. First, a single function — disinfection — governs the public-health consequence. That function can be defeated in two directions: under-dosing, which admits pathogens, and over-dosing, which distributes a chemical hazard in finished water. Second, the plant holds a bulk inventory of a hazardous disinfection chemical (chlorine gas, sodium hypochlorite, or an equivalent) whose storage is at once the disinfection supply and a chemical hazard to the surrounding area. Third, the plant's control surface — dosing setpoints, chemical-feed pump controls, residual analyzers, and the indication on which operators rely — is the locus of the cyber design basis. A single governing function, a dual-direction chemical hazard, and a consequence-bearing control surface: these three define the class and distinguish it from every other class in the series.

2.2 The Governing Configuration

The conventional surface-water plant is the governing configuration because it is the bounding case on both consequence chains developed in Section 3: it carries the largest chemical inventory and treats source water whose quality is variable and can itself be attacked. Requirements stated

for the class are validated against this configuration. The groundwater plant is treated as a proper subset, and Section 8 records which class requirements it inherits unchanged and which fall away with the absent unit processes.

2.3 Deployment Typologies and Size Range

The class spans surface-water and groundwater plants and the full size range, from large regional facilities serving hundreds of thousands to small community systems, including the sub-threshold tail serving populations below the America's Water Infrastructure Act reporting line that the sector framework recognizes at framework §1. Class requirements are stated for the governing configuration; the bounded minimum-requirement set for sub-threshold plants is a forthcoming DBA-W instrument (framework §1; reference event DBA-W-RE-1 open item), and this instrument does not attempt to substitute for it.

2.4 Consequence Context

Treatment-plant outage is the governing consequence event of the water sector because loss of disinfection converts a supply interruption into a public-health emergency within hours rather than days. The time-to-harm is set by the disinfection residual already in the distribution system and by the finished-water storage buffer, not by the plant's return-to-service schedule. The class inherits the framework's mapping of the four critical functions to federal Emergency Support Function activation thresholds (framework §4.3); a treatment plant that holds the class adequacy result is simultaneously reducing the federal response burden its failure would impose. The plant's consequence is not water quality alone: sustained supply is also what provides firefighting flow, so a treatment or supply loss is at the same time a loss of fire-protection capability — a consequence the class weights alongside potable-water safety.

3. The Facility-Class Design Envelope

3.1 Why the Treatment Plant Is the Governing Class

The sector framework places the treatment plant first among facility classes because treatment outage is the governing consequence event (framework §10.2). This instrument does not re-derive that placement; it develops the envelope the placement implies. The governing envelope is set by the two consequence chains in Section 3.2, which no other facility class carries together.

3.2 The Two Consequence Chains

The treatment-plant class is distinguished by the fact that a single initiating event can open two independent consequence chains.

Chain A — loss of disinfection. Manipulation, failure, or supply interruption drives the disinfectant residual outside the safe band — below the level that controls pathogens, or above the level that is chemically safe — and unsafe water is distributed. The chain runs through the water to the served population and activates on the timescale of distribution-system hydraulics.

Chain B — chemical release. A precision strike, catastrophic mechanical failure, or deliberate sabotage of bulk chlorine or hypochlorite storage releases a toxic quantity to the surrounding area. The chain runs through the air to the population around the plant, on the timescale of a release plume, and it eliminates the disinfection supply at the same time.

A single strike on chemical storage therefore initiates both chains at once — a public-health chain through the water and a chemical-hazard chain through the air — from one action. The design basis must carry both. This is the feature that distinguishes FC1 from the other facility classes and requires a dedicated chemical-release event category (Section 4.2), which the sector framework's general event categories, oriented to the water pathway, do not provide.

3.3 Design Parameters at Class Resolution

Three parameters govern the envelope: the disinfection residual (the regulatory minimum is the floor; the class parameter is the margin above it and the time to breach that margin under each event category), the on-site chemical inventory (the class parameter is the sustainable treatment duration without resupply and the hardening standard for its storage), and the finished-water storage buffer (the time within which recovery must occur before the consequence reaches the population). This instrument states the parameters and their governing logic. Their quantitative values are pending validation with water-utility operators and state primacy agencies (Section 9) and are not asserted here as fixed numbers, consistent with the development sequence the sector framework specifies for FC1 (framework §10.4).

4. Design Basis Events: Non-Adversarial Envelope

The following event categories are the facility-class instantiation of the sector framework's general event categories (framework §4.2), together with one category — chemical release — that is specific to this class. Each is stated in the framework's structure of postulated event, consequence threshold, required function, and design-basis duration. Requirements the sector framework already establishes are inherited, not repeated; this section states their application to the treatment plant.

4.1 Loss of Disinfection Capacity

Postulated event: failure, manipulation, or supply interruption of the disinfection process such that the delivered residual falls outside the safe band.
Consequence threshold: distribution of water with residual below the pathogen-control minimum, or above the chemical-safety maximum, to any portion of the served population.
Required function: the dosing system must fail to a conservative safe condition, and an out-of-band setpoint must not reach the process without a positive local confirmation that cannot be executed remotely (inherits framework §5.2 interlock independence and the indication-integrity requirement).
Design-basis duration: indefinite for the cyber and equipment-failure vectors; supply-limited for the chemical-supply vector.

4.2 Chemical Release from Storage and Feed Systems

Postulated event: a precision strike, catastrophic mechanical failure, or deliberate sabotage of bulk chlorine, sodium hypochlorite, or anhydrous ammonia storage or feed equipment (anhydrous ammonia, used for chloramination, is the more hazardous release from an exposure standpoint), releasing a toxic quantity to the surrounding area.
Consequence threshold: an airborne chemical concentration hazardous to the population surrounding the plant, together with the concurrent loss of the disinfection supply.
Required function: chemical storage must be sited, hardened, and instrumented so that a single credible strike does not simultaneously breach containment and eliminate the disinfection supply; secondary containment, gas detection, isolation, and an emergency gas scrubber are required, and a backup disinfection pathway independent of the primary storage must exist.
Design-basis duration: event-specific for the release; supply-limited for the associated loss of disinfection.
This is the dedicated second-chain category the sector framework requires FC1 to carry (framework §10.2). It interfaces with chemical-safety regimes the water design basis does not otherwise reach (Section 7).

4.3 Cyber Attack on Treatment Operational Technology

Postulated event: intrusion into the treatment control system producing manipulation of a dosing setpoint, modification of the controller program or logic, or falsification of the indication presented to operators.
Consequence threshold: out-of-band dosing distributed to the population, or loss of authentic operator visibility during a treatment excursion.
Required function: the treatment control surface inherits framework §4.2.2 and §5.2 as revised in v1.6 — interlock independence, credential architecture, monitoring, indication integrity, and protected engineering objects — applied specifically to the dosing setpoint, the chemical-feed pump control, the residual analyzer indication, and the engineering workstation, vendor engineering software, and controller project file for this plant.
Design-basis duration: indefinite (framework §4.2.2).
This category is characterized empirically by two reference events issued under the sector framework: DBA-W-RE-1 (the coordinated cyber attack on Minnesota community water systems, 2026) and DBA-W-RE-2 (the Oldsmar water treatment plant event, 2021). RE-1 establishes the falsified-indication and common-substrate findings this category inherits; RE-2 establishes that the design-basis reading is independent of attribution.

4.4 Loss of Electric Power to Treatment

Postulated event: loss of grid-supplied power to the treatment plant beyond emergency-generation capacity or consumable limits.
Consequence threshold: loss of treatment or disinfection beyond the finished-water buffer.
Required function: on-site stored energy, emergency generation, or passive means sufficient to sustain disinfection for the class duration (inherits framework §4.2.1)
Design-basis duration: 30 days, the class default (framework §4.2.1).

4.5 Loss of Chemical Supply

Postulated event: disruption of the treatment-chemical supply chain for a duration that depletes on-site inventory.
Consequence threshold: inability to maintain disinfection residual at regulatory minimums beyond the on-site inventory duration.
Required function: on-site inventory and secure storage sufficient to sustain treatment for the class duration, or a demonstrated alternate disinfection pathway (inherits framework §4.2.5).
Design-basis duration: 30 days (framework §4.2.5).

4.6 Source-Water Contamination and Physical Attack on Critical Nodes

Postulated event: contamination of source water, or physical attack on the intake, the clearwell, or the filtration works, that bypasses or overwhelms treatment.
Consequence threshold: distribution of inadequately treated water, or a single-point loss of treatment capacity.
Required function: monitoring and detection capable of identifying contamination before distribution, and hardening of single-point-of-failure nodes (components whose individual loss defeats treatment), and their protection by trains that are robust, diverse and independent — two or more trains, differing in kind, and sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all (framework §4.1); redundancy alone, or diverse trains that share a room, are not credited against deliberate attack (DBA-MA-DiD §3.4) (inherits framework §4.2.3 and §4.2.4).
Design-basis duration: event-specific for the contamination limb (framework §4.2.4); 30 days for full restoration and 24 hours for minimum service to critical dependent facilities for the physical-attack limb (framework §4.2.3). (These minimum-service intervals are inherited from the framework and remain among the quantitative parameters flagged for operator and state-primacy validation in Section 9.)

4.7 Natural-Hazard Exceedance

Postulated event: a natural-hazard event exceeding the design parameters of one or more critical plant facilities (inherits framework §4.2.6).
Consequence threshold and required function are inherited from the sector framework; the class adds only that chemical storage must remain within its containment design basis under the exceedance event.
Design-basis duration: 30 days for full restoration; 72 hours for minimum service (framework §4.2.6).

Table 4-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Inherits / adds
Loss of disinfection	Residual driven outside the safe band by failure, manipulation, or supply loss	Chain A (water)	framework §4.2.2, §4.2.5
Chemical release	Strike, failure, or sabotage of bulk chlorine/hypochlorite storage	Chain A + Chain B (air)	adds (class-specific)
Cyber attack on OT	Dosing manipulation, program/logic modification, falsified indication	Chain A (water)	framework §4.2.2, §5.2 (v1.6)

Event category	Postulated event (class-specific)	Consequence chain	Inherits / adds
Loss of electric power	Grid loss beyond generation/consumables	Chain A (water)	framework §4.2.1
Loss of chemical supply	Supply-chain disruption depleting inventory	Chain A (water)	framework §4.2.5
Source-water / physical	Contamination or strike on intake, clearwell, filtration	Chain A (water)	framework §4.2.3, §4.2.4
Natural-hazard exceedance	Hazard beyond facility design parameters	Chain A + containment	framework §4.2.6

4A. The Adversarial Design Basis for the Facility Class

The sector framework's three-tier threat-actor taxonomy (framework §7.5.1) assigns to this class as follows. Tier 3, the sophisticated non-state actor, is the class's non-adversarial cyber baseline — the Oldsmar and Minnesota reference events sit here — and is addressed by the Section 4 requirements. Tier 1 (state) and Tier 2 (near-state proxy) trigger the military-action extension of Section 5 — the state-actor reading most people bring to that term, made explicit here. The class adopts the v1.6 refinement that tier denotes direction, not a ceiling on capability: a Tier 3 actor may wield proliferated state-grade tools. Hence, the class assesses the capability an actor can bring to the dosing control surface and the chemical storage independently of the actor's tier. The governing adversarial event for the class is the coordinated cyber compromise of the dosing control surface with falsified indication; under the military-action extension, it is joined by the precision strike on chemical storage.

5. Military-Action Extension: DBA-MA for the Water Treatment Plant

5.1 MA Extension Applicability Determination

Every treatment plant performs the facility-level military-action applicability determination the sector framework makes mandatory (framework §7.1): the primary factor is geolocation relative to military-action exposure, and the secondary factor is strategic value, with the determination time-varying and subject to the framework's re-determination triggers. For this class, a plant whose loss would remove potable water from a population or a strategic node above the framework threshold, or a plant within military-action exposure, carries the extension. The determination is recorded, re-evaluated against the framework triggers, and its scope reductions carry a recorded basis (consistent with Section 5A.5).

5.2 The Four Characteristics Applied to the Class

Applied to the treatment plant, the four military-action characteristics take specific form. Intentionality: the adversary selects the dosing setpoint or the chemical-storage tank as the target. Persistence: the adversary pre-positions within the treatment SCADA and holds access across the interval needed to time the act. Precision: the adversary targets the specific chemical, setpoint, or

storage vessel whose defeat produces the governing consequence. Denial: the adversary falsifies indication and times the act to minimum staffing and concurrent stress, denying the operator both the accurate picture and the response margin. The class validation standard is the state actor who has studied this plant's architecture, identified its weakest point, and designed the act to defeat every detection mechanism.

5.3 The Chemical-Storage Strike as the Governing MA Scenario

The governing military-action scenario unique to this class is the precision kinetic strike on bulk chlorine or hypochlorite storage. It realizes both consequence chains from a single action: a toxic release to the surrounding population and the simultaneous loss of disinfection supply. The extension therefore requires that chemical storage be hardened and given standoff appropriate to the plant's applicability determination, and that a disinfection pathway survive the loss of the primary storage. This scenario has no analog in the non-adversarial envelope and is the principal reason the class carries a dedicated chemical-release event category.

5.4 The Coordinated Cyber Scenario

The governing military-action cyber scenario is the pre-positioned SCADA compromise that manipulates dosing through a path requiring no operator-visible action, falsifies the indication the operator would rely on, and is timed to grid stress and minimum staffing — the state-actor escalation of the Oldsmar event that DBA-W-RE-1 characterizes at sector scale. Against this scenario, the class relies on the indication-integrity and protected-engineering-objects requirements (Section 4.3) and on the autonomous safe condition of Section 5A, because operator detection cannot be assumed and displayed values cannot be trusted.

5.5 Cross-Sector Interaction

The treatment plant's military-action envelope is coupled to other sectors through the water-electricity interdependency (framework §8.1), which the adversary may exploit by timing a cyber or kinetic act to a concurrent loss of grid power, and through the treatment-chemical supply chain, whose disruption is both an independent event category (Section 4.5) and a force multiplier for the chemical-storage scenario. The finished-water interface to distribution (FC2 and FC3) sets the recovery-time boundary condition and is treated as an external dependency of this class. Where this plant is the sole cooling-water supply path for a thermoelectric generation facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-ES series, that supply relationship carries tightened acceptance criteria, which discharges for this class the reverse-direction obligation DBA-EN-SF §5.4 places on the DBA-W series (framework §8.1). The criteria are quantified with the operator of the generation facility; who bears the cost of meeting them is open and is not settled by this instrument.

5A. The Adequacy Result: Autonomous Safe Treatment Condition

5A.1 The Autonomous Safe Treatment Condition (ASTC)

The class adequacy result is the Autonomous Safe Treatment Condition: the condition in which, on loss of networked control and external support, the plant holds a conservative safe state autonomously. Dosing fails to a safe condition rather than to an unsafe one; distribution isolation is manually operable at the valve without SCADA; and a boil-water fallback preserves the public-health boundary where residual integrity cannot be confirmed. The ASTC must be achievable and sustainable by a minimal on-site crew for a defined duration without external communications, inheriting the local-sovereignty principle of framework §9.4.

5A.2 Adequacy in the Non-Adversarial Domain

In the non-adversarial domain, the ASTC is achievable with the Section 4 requirements: fail-safe dosing, interlock independence, on-site power and chemical inventory for the class duration, and manual procedures maintained by trained operators. The class is adequate against the non-adversarial envelope where these requirements are met and demonstrated.

5A.3 Adequacy in the Adversarial Domain, Bounded

In the adversarial domain, adequacy is bounded rather than unconditional. Against the cyber chain, indication integrity, protected engineering objects, and offline gold-copy recovery bound the event and preserve the ASTC even when displayed values are falsified. Against the chemical-release chain, storage hardening, standoff, secondary containment, and an independent backup disinfection pathway bound the event. Adequacy holds within these bounds; beyond them, the residual of Section 5A.4 governs.

5A.4 The Declared Residual

Two residuals are declared rather than concealed. First, a sufficiently large precision strike on chemical storage can realize the chemical-release chain regardless of the plant's cyber posture; hardening and standoff reduce but do not eliminate it, and beyond the design strike it is a declared residual driving standoff siting and emergency planning. Second, sustained denial of site access can outlast on-site chemical inventory and generation; beyond the class duration the residual is a loss of treatment that the recovery architecture, not the plant's autonomy, must address. These residuals set the boundary of the class adequacy claim and are the inputs to the military-action hardening decisions of Section 5. They are the declared residuals of the DBA-MA-DiD §3.5 adequacy statement, which framework §4.1 makes owed.

5A.5 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a reduced chemical inventory duration, a reduced finished-water buffer, a reduced standoff — must be justified against the class baseline under the consequence-basis conditionality principle the framework makes mandatory for every facility-class instrument. The scaling corollary applies: the larger the claimed relaxation relative

to the conservative baseline, the stronger the analytical basis required to support it. A relaxation without a recorded basis is not consistent with the class design basis.

6. The Dual-DB, Stated in Full

6.1 The Two Design Bases

The treatment plant carries two design bases simultaneously (framework §10.1): a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here. The two are not independent; they interact at the point developed in Section 6.4.

6.2 The Physical-DB Half: The Determinism Posture

The physical design basis is the set of protective functions the plant relies on to hold its safe state independently of the networked control system: fail-safe dosing, physical interlock independence, the integrity of indication, and the containment of chemical storage. Those among them that decide on measured quantities against setpoints are to be locally sovereign as FD-LS §3.2 defines the property: the decision is physical law, the function is colocated with the plant, and no authority is reachable in either limb, so that nothing on the reachable side can withhold, delay or induce its action, and nothing there can alter its setpoints, logic or bypass state except through a governed act. FD-LS governs and is not restated here. Above the Bright Line — the boundary drawn on the consequence of each action — a component within scope of FD-LD §4.3 may propose but may not execute without command-broker authorization. The determinism posture is a scope posture, not a placement. The disinfection dosing loop, the chemical-feed control, and the safety interlocks, where deterministic within a stated input bound and formally testable as deployed, meet the carve-out and are outside the Command Broker obligation for their own in-bound actions (FD-LD §4.3). They are not below the line: they take some of the highest-consequence actions at the plant, and they are outside the obligation because their behavior is bounded by proof (FD-BL §4.5). Any component that cannot be shown to meet the carve-out is within scope; each of its actions is placed on its own consequence (FD-BL §4.2), and under the conservative implementation of framework §5.5 each action it would carry into the physical command layer is treated as above the line unless a documented per-action assessment places it below. A change an in-scope component proposes to a protective function’s setpoints, logic, bypass state or actuation criteria is an action upon that function, placed on its own consequence (FD-BL §4.4).

6.3 The AI-Governance-DB Half: Named and Referenced

Machine-learning and generative-AI systems in treatment control — dosing optimization, predictive maintenance, and operator decision support — are governed by the Command Broker and Bright Line architecture applied at framework §5.5 and established cross-sector in the Command Broker family (CB-Framework_Cross_Sector, with CB-Annex-Nuclear as the sector binding case; CB-REG, the Command Broker Family Register, is authoritative for version state).

This instrument does not redefine that architecture; it names it and assigns to it the plant's highest-criticality parameter, the disinfection dosing loop. Placement is by the consequence of each action, not by the component (FD-BL §4.2, §4.3). An anomaly-detection model on a historian that meets the carve-out is outside the Command Broker obligation for its in-bound actions; that is not a grant of ungoverned autonomy and not a finding that its actions are below the line (FD-LD §4.3). A generative assistant is within scope: a dosing change it proposes, together with an operator’s acceptance of it, is one action placed on the consequence of the change, which for the disinfection dosing loop is above the line and requires command-broker authorization; its output that would make no change of its own is placed below the line and carries the indication-integrity obligation of FD-BL-D2 wherever it can ground an authorization.

6.4 The Interaction Point: ASTC Under Denial

The two design bases meet at the Autonomous Safe Treatment Condition under denial. The control system must be architecturally incapable of executing an above-line action originating from a component within scope of FD-LD §4.3 — including any command that moves a highest-criticality treatment parameter — without command-broker authorization, and the ASTC must hold with no reachable authority of any kind. The physical design basis guarantees the safe state; the AI-governance design basis guarantees that no proposing system can move the plant out of it without a qualified human authorization the architecture can enforce.

6.5 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is FD-BL with its determinations, and the Command Broker family under CB-REG. Where the two are cited together, the physical requirement governs the plant's protective architecture and the AI-governance requirement governs the authorization of any proposing system that interacts with it.

7. Regulatory and Institutional Context

The treatment-plant class sits under the Safe Drinking Water Act and its disinfection and residual rules, administered through state primacy agencies that hold the mandate authority the sector otherwise lacks. America's Water Infrastructure Act, which amended the Safe Drinking Water Act, adds risk-and-resilience and emergency-response-plan obligations that are the near-term propagation vehicle for the class requirements, and engineering practice (framework §2.5) is the mechanism by which the class design basis reaches plants ahead of any mandate. The Environmental Protection Agency's Office of Water and the Cybersecurity and Infrastructure Security Agency issue the water-sector cyber guidance the class inherits, including the advisory associated with the Oldsmar event.

The chemical-release event category interfaces with regimes the sector framework does not reach: the Environmental Protection Agency's Risk Management Program for chlorine and other regulated substances, and the Occupational Safety and Health Administration's Process Safety

Management standard. The class treats these as the governing authorities for the chemical-hazard chain. It requires that the plant's Risk Management Program and Process Safety Management analyses be consistent with the chemical-release event category of Section 4.2.

8. Inheritance Map and Anticipated Derivatives

8.1 The Inheritance Map

The sector framework requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally (framework §10.1). Table 8-1 is that map for this class.

Table 8-1. Inheritance map for DBA-W-FC1.

Element	Source in framework v1.6	Inheritance mode
Threat-actor taxonomy	§7.5.1	Inherited unchanged
Design-basis principles / bounding method	§4.1	Inherited unchanged
Cyber requirements (interlock, credential, monitoring, indication integrity, protected engineering objects)	§5.2	Inherited unchanged, applied to the treatment control surface
Command Broker / Bright Line architecture	§5.5; Command Broker family (CB-REG)	Inherited unchanged, referenced
Event categories	§4.2	Inherited with adaptation; adds the chemical-release category
MA-applicability determination	§7.1	Inherited with adaptation to facility level
Adequacy standard / local sovereignty	§9.4	Inherited with adaptation as the ASTC
Consequence-basis conditionality check	framework portfolio requirement	Inherited with adaptation
The two consequence chains and chemical-release category	§10.2 (scope)	Locally developed
Class design parameters and the ASTC	—	Locally defined

8.2 Anticipated Derivatives

Site-specific, state-specific, and contract-specific treatment-plant design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. Technical depth belongs in this class document; derivatives stay bounded and read from it. A facility-class use case (DBA-W-UC-1), tracing the full consequence-chain of an area-denial attack on a large surface-water treatment plant, is the next instrument anticipated under this class.

9. Gaps and Open Items

9.1 Quantitative Design Parameters — Pending Operator and State-Primacy Validation

The residual margins, chemical-inventory durations, and finished-water storage-to-population buffers named in Section 3.3 are stated as parameters, not as numbers. The sector framework specifies that FC1 development requires engagement with water-utility operators and state primacy agencies to validate the event categories and adequacy standards against real facility architectures (framework §10.4); the quantitative values await that engagement.

9.2 Chemical-Release Thresholds and Standoff Standards

The quantitative release thresholds, standoff distances, and hardening standards for the chemical-release category (Section 4.2) are open. They must be developed against the Risk Management Program and Process Safety Management analyses named in Section 7. This is the most class-specific open item and has no counterpart in the sector framework.

9.3 Sub-Threshold Minimum-Requirement Set

The bounded minimum-requirement set for sub-threshold treatment plants (Section 2.3) is a forthcoming DBA-W instrument, not developed here. Its relationship to this class — which requirements bind below the reporting threshold and which are relaxed — is an open item linked to the sector framework §1 scope statement and reference event DBA-W-RE-1.

9.4 Groundwater-Variant Scope Reduction

The specific class requirements that fall away for the disinfection-primary groundwater variant (Section 2.2) are recorded in principle in the inheritance map but require validation against representative groundwater plant architectures before the reduction is fixed.

9.5 Facility-Class Use Case

DBA-W-UC-1, the area-denial use case that would demonstrate the full consequence-chain trace for this class, is anticipated but not yet developed. It is the vehicle for exercising the two consequence chains against a worked scenario and is the recommended next instrument under this class.

References

DBA-W Series and Parent Framework

DBA-W-Sector_Framework v1.14. Design Basis for Water Sector Infrastructure. Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued and from which it inherits (Section 8).

DBA-W-RE-1 v0.7 and DBA-W-RE-2 v0.2. Reference Event Instruments (Minnesota community water systems, 2026; Oldsmar water treatment plant, 2021). Eclectic Technologies, 2026. Empirical basis for the cyber event category (Section 4.3).

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics and the design-basis-under-military-action methodology.

Command Broker Framework Series

CB-Framework_Cross_Sector and CB-Annex-Nuclear. Eclectic Technologies, 2026. Cited by designation; version state in CB-REG, the Command Broker Family Register. Governing instruments for the AI-governance design basis named and referenced at Section 6.3.

Regulatory and Standards

Safe Drinking Water Act; America's Water Infrastructure Act (2018) and its 2023 Operational Guide; EPA Risk Management Program; OSHA Process Safety Management standard; CISA/EPA/FBI Joint Cybersecurity Advisory AA21-042A (2021).

Revision History

Version	Date	Author	Description
0.1 — DRAFT	August 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-W series: the water treatment plant. Establishes the two consequence chains and the dedicated chemical-release event category (§3–§4), the facility-class event envelope, the adversarial and military-action extensions (§4A–§5), the Autonomous Safe Treatment Condition adequacy result and declared residuals (§5A), the dual-DB stated in full (§6), the regulatory context including the RMP/PSM chemical overlay (§7), and the required inheritance map (§8). Inherits DBA-W-Sector_Framework v1.6. Quantitative design parameters flagged pending operator and state-primacy validation (§9).
0.2	August 2026	Timothy E. Roxey	Section 4 event structure reformatted so each field — postulated event, consequence threshold, required function, design-basis duration — sits on its own bolded line. Inherited design-basis durations stated inline with their framework references (e.g., 30 days, framework §4.2.1); prose otherwise unchanged.
0.3	August 2026	Tim Roxey	West Yost review (A. Ohrt) — clear edits: fire-flow added as a consequence (§2.4); anhydrous ammonia and an emergency gas scrubber added to the chemical-release event (§4.2); 'most complex control surface'

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			rationale removed (§2.2); AWIA stated as amending the SDWA (§7); single-point-of-failure clarified (§4.6); minimum-service durations noted as framework-inherited and open (§4.6, §9); tier-vs-military-action reading clarified (§4A); clearwell/contact-storage stated as in-plant at the FC1/FC2 boundary (§1.2). Governing-function reframe (beyond disinfection) held for author decision.
0.4	September 2026	Tim Roxey	FD citation form conformed under the citation-form determination. The governing-instruments paragraph cited the Foundational Definitions with version tokens, two of which had gone stale on 3 September (FD-BL-D1 re-issued at v0.4, FD-BL-D2 at v0.3) — a version-pinned citation resolving into superseded text. Tokens dropped; the family is cited by designation and resolves to FD-REG. No requirement, event category, parameter or finding changed.
0.5	October 2026	Tim Roxey	Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026 (P9-07), together with the acts OI-14 placed at this revision. §6.2 no longer places the protective functions below the line as deterministic. The determinism posture is stated as scope: the functions are outside the Command Broker obligation by the carve-out and not by placement (FD-LD §4.3; FD-BL §4.5), and a component not shown to meet the carve-out is within scope with each action placed on its consequence, the conservative implementation of framework §5.5 keeping the outcome the earlier text reached. Its local-sovereignty sentence is read as FD-LS's construct and cites FD-LS §3.2 with both limbs and the governed act. §6.3 places by the consequence of each action rather than by component and no longer grants the historian model autonomy (OI-14). §6.4

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			states the obligation's reach by origin in a component within scope of FD-LD §4.3 rather than as AI-originated (OI-14). Whether each outcome so restated is right on consequence is the author's to confirm. The Command Broker family is cited by designation against CB-REG rather than by project at §6.1, §6.3, §6.5, the inheritance map and References (OI-2, OI-4). The parent framework is cited at v1.13 at the issued-under and parent-document lines, §1.4 and References (OI-3); provenance notes recording a value established at v1.6 stay as written. Reference-event pins are repaired to current. The required function for single-point-loss nodes credits independent protection, with redundancy kept beside it against random loss and not in its place, and the declared residuals of §5A.4 are identified as the DBA-MA-DiD §3.5 output that framework §4.1 now makes owed (OI-11). OI-12, the reverse-direction cooling-water rule of DBA-EN-SF §5.4, is not taken at this revision and stays open for the author. Held, not missed: the local-sovereignty principle cited from framework §9.4 at §5A.1 was read and left as written, pending Project 5's answer on the DBA-MA construct. Revision-history rows were excluded from substitution.
0.6	October 2026	Tim Roxey	Two author decisions of 3 October 2026. The required function for single-point-loss nodes credits trains only when they are robust, diverse and independent, sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all (framework §4.1); redundancy alone, or diverse trains that share a room, are not credited against deliberate attack. Section 5.5 carries, conditionally, the reverse-direction cooling-water obligation of DBA-EN-SF §5.4: where

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			the facility is the sole cooling-water supply path for a Tier 2 or Tier 3 DBA-ES thermoelectric facility, that supply relationship carries tightened acceptance criteria, quantified with the generation facility's operator, and who bears their cost is left open (OI-12 closed). The parent framework is cited at v1.14 and adjacent-class pins are repaired to current. Revision-history rows were excluded from substitution.